

Demonstration of Meta-antenna: Mechanically Frequency Reconfigurable Metamaterial Antennas

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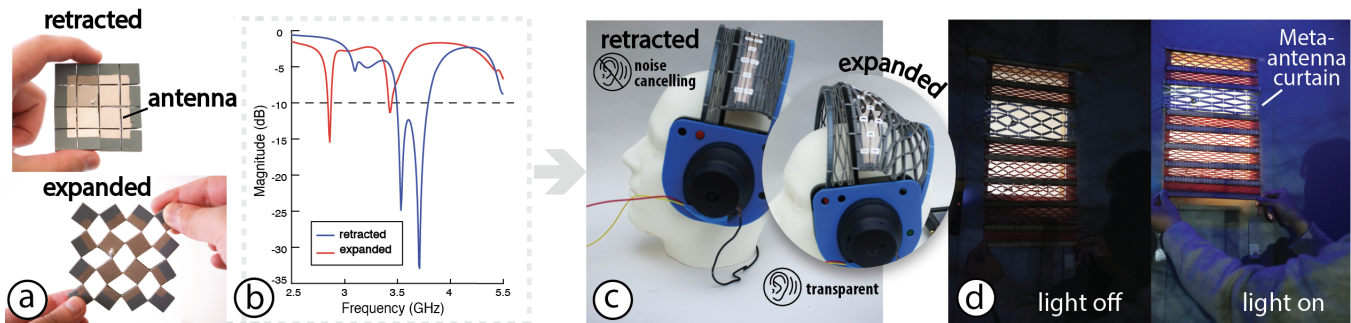


Figure 1: a) Meta-antennas are mechanically reconfigurable metamaterial antennas that dynamically expand and retract. (b) As Meta-antennas deform, their frequency response shifts based on their deformed states. Meta-antennas can be used to construct (c) an adaptable headset that alters modes between noise-cancelling and transparent, and a (d) smart curtain that adjusts its lighting based on the antenna's expanded state.

Abstract

In this demo, we present Meta-antenna, a design and fabrication approach for building frequency-reconfigurable antennas using a single type of mechanical metamaterial structure. Unlike conventional antennas with fixed geometries and static frequency and radiation properties, Meta-antenna employs mechanical reconfiguration to dynamically adjust both its geometry and radiation behavior, enabling greater adaptability for sensing and communication applications. We demonstrate Meta-antenna's potential through

adaptive personal devices, smart home systems, and tangible user interfaces.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

metamaterial mechanisms, reconfigurable antenna, wireless communication and sensing

*Co-last authors, equal contribution.

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1 Introduction

Antennas are essential to many technologies we depend on daily—from the high-speed WiFi antennas in our smartphones to the satellite dishes delivering our favorite TV shows.

Recently, the HCI community has explored ways to incorporate antennas into interactive systems. These systems support applications such as gesture recognition, energy transfer, and mechanical motion sensing. For instance, researchers have demonstrated how antennas can recognize human gestures and translate them into an avatar's movements in XR [8, 9], can be integrated into wearable devices for energy harvesting [15], and can be used as encoders for mechanical motion tracking [7]. In all these applications, the effectiveness and functionality of the antennas rely heavily on their radiation efficiency, which is influenced by various factors, including the antenna's geometry [13], material [16], and environment of operation [5].

Traditionally, antennas are designed with *fixed geometries* corresponding to fixed Radiofrequency (RF) properties. For example, the patch antenna designs in PrintedWifi [7] track mechanical rotations, yet remain static antennas with fixed resonant frequencies and radiation patterns. While fixed geometry antennas work well for certain applications, they are less effective for broader use-cases, such as environment-adaptive sensing [17], mode switching between data transmission and reception [2], and development of dynamic antennas that seamlessly operate across a range of network protocols [11]. This limitation in static antenna capabilities highlights the need to design *dynamic* antennas that can change their RF properties.

To overcome these limitations, researchers have developed reconfigurable antennas that can change their RF parameters while adopting a single antenna structure. One approach is through mechanical reconfiguration which relies on the antenna's geometry being physically altered. Such mechanically reconfigurable systems utilize actuators, like motors, to move assemblies of linkages [12], gears [12, 14] and ground planes [3].

However, using actuator-based mechanical methods can exponentially increase the mechanical complexity of reconfigurable antenna systems when multiple motion primitives such as bending, expansion, and rotation are required.

To address these limitations, we propose a mechanical building block structure that enables these motion primitives, while ensuring simplified designs for mechanically reconfigurable antennas.

Mechanical metamaterials are one example of building block structures that have been explored for interactivity within HCI [4, 6]. These metamaterials can be programmed to take on various shapes [10] and support multiple primitive motions such as bending, expansion, and rotation. Researchers have even utilized these meta-material structures to construct mechanically reconfigurable antennas instead of using the need for bulky mechanical assemblies. Existing work has primarily explored how incremental compressions for a single mechanical metamaterial structure affect resonant frequency and radiation pattern, without exploring how changes in geometry or other interactions, like rotation, influence antenna behavior. [1, 18]. In other words, existing metamaterial

antenna systems do not fully exploit the capabilities of metamaterials. Through reversible geometric transformations, metamaterial structures have the potential to serve as reconfigurable antennas that adapt to a range of interaction scenarios. As a result, the design and fabrication of programmable metamaterial antennas—and their application to fundamental interactive behaviors—remain largely underexplored.

In summary, we contribute:

- *Meta-antenna*, a mechanically reconfigurable metamaterial antenna geometry that acts as a building block for reconfigurable antenna systems
- an Ansys-based Meta-antenna construction and simulation tool with a Python-Arduino code template for sensing interactions
- a fabrication pipeline for constructing mechanically and electrically durable Meta-antenna structures
- five technical evaluations on the electrical durability and fabrication method of Meta-antenna
- three applications for smart home systems, adaptive electronic devices and tangible user interfaces.

2 Meta-antenna Fabrication

We fabricate our Meta-antennas by laser cutting the metamaterial geometry, followed by applying conductive spray paint to form the ground and patch layers.

2.1 Laser Cutting Metamaterial Geometry and Patch Antenna Mask



Figure 2: (a) Meta-antenna geometries are fabricated from lasercut rubber sheets. (b-c) the rubber sheets are masked with painter's tape that is lasercut to the antenna's patch pattern shape.

Laser-cutting of Metamaterial Geometry and SMA Probe Location:

We use laser-cutting as our main fabrication method. We laser cut off-the-shelf laser-cuttable rubber sheets for the main metamaterial geometry. On our Universal laser cutter (PLS6.150D), we laser cut the rubber sheets with 100% power, 3.3% speed, and 200 ppi. We leave the laser cut rubber sheet inside the laser cutter for the next fabrication step. Besides cutting out the metamaterial geometry, this step also cuts out a hole for the SMA probe.

Laser-cutting of Antenna Masks: To create the antenna patch pattern using conductive spray paint, we laser cut a masking layer that covers areas not meant to be spray-painted. To accomplish this, we cover the lasercut Meta-antenna geometry, while still inside the laser-cutter, with painter's tape. We laser cut the painter's tape mask with 16% power, 100% speed, and 500 ppi. After laser cutting

the mask, we remove the Meta-antenna geometry from the laser cutter.

2.2 Spraypainting Grounding Layer and Patch Antenna

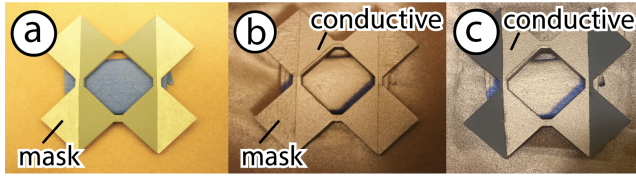


Figure 3: Meta-antenna patches are fabricated with conductive spray paint.

Spray-painting the Grounding Layer: Next, we spray-paint the entire bottom of the metamaterial geometry with a copper-based conductive material using an airbrush (Master G233). We use MG-chemicals silver coated copper spray paint (843WB), which is highly conductive (5.3×10^{-4} resistivity). To ensure good coverage, we apply two layers (0.01778mm) of the spray paint, waiting 30 minutes between each layer to air dry.

Spray-painting the Conductive Patch Pattern: Before applying the conductive spray paint to form the patch on top of the metamaterial geometry, we first remove the sections of masking tape covering the areas intended for conduction. We then spray-paint the exposed regions, following the same procedure used for the ground layer. After allowing the second conductive layer to dry for 30 minutes, we peel off the remaining masking tape from the rest of the structure.

2.3 Coating to Improve Electrical Durability at the Hinges

Coating the Meta-antenna: In the final step, we coat the hinges only of the Meta-antenna top and bottom surfaces with a flexible acrylic paint (PlaidFX Flexible Acrylic Paint) which can either be spray-painted or directly applied via paintbrush. We found that applying the coating improved the electrical connection durability of the Meta-antenna at the hinges (see Section ??). In fact, without applying the flexible coat, the Meta-antenna structure electrically breaks after only one compression. After coating, we let the Meta-antenna air dry for 1 hour before using it.

2.4 Meta-antenna Assembly

After fabrication, we connect our Meta-antenna structure to its supporting electronic components.

Connecting SMA Probe to MetaAntenna Geometry: We choose a coaxial SMA probe (TAOGLAS CAB.058) that has a 50Ω impedance. We insert the SMA probe into the corresponding hole on our Meta-antenna. To secure the SMA probe in place, we use a two-part silver conductive epoxy (MG Chemicals 8331D) and apply it to the SMA probe location on the grounding and patch layers. We then cure the epoxy in an oven (70°C for 15 minutes).

3 Applications

Our technique enables users to weave antennas into dynamic everyday life objects like responsive furniture for smart living, adaptive personal electronic devices, and interactive tangible user interfaces.

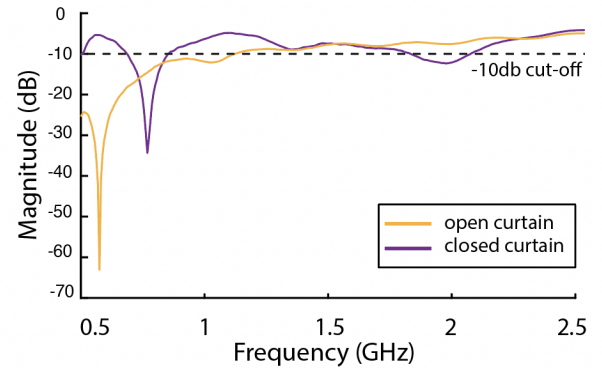
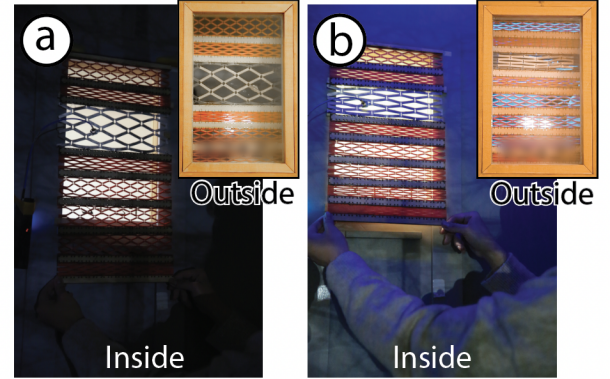


Figure 4: This smart light-adjusting curtain dims as it expands

3.1 Responsive Furniture for Smart Living

Modern architecture has explored shape-morphing furniture and structures more widely in recent years. In order to enable smarter and more responsive living spaces, embedding these structures with the capability to detect what state they are makes these structures even more compelling.

We demonstrate this notion of responsive architecture with an example of a curtain that undergoes shape change. In this application, we demonstrate how Meta-antennas can transform a curtain into a smart piece of furniture that adjusts household lighting based on its expanded (holes) and retracted (no holes) states. By leveraging the inherent kirigami design of our Meta-antenna, which has holes in its expanded state and no holes in its retracted state, we map the expanded state to dim the room lights as it allows natural light to enter. In contrast, the retracted state, with no holes, increases the room brightness. We constructed our metamaterial expandable curtain with a middle segment functioning as a meta-antenna. The main interaction primitive we showcase in this example is expansion/retraction. As shown in Figure 4, when the curtain expands by

133.33%, the resonance frequency shifts by **0.194GHz (194 MHz)**, which is a PCRf (Percentage Change Resonant Frequency) value of **25.24%**. This shift in frequency is detected by our VNA and results in adjusting the LED strip's brightness. As illustrated in Figure 4 (a-b), when the curtain is expanded, outside lighting enters the room, thus the LED dims out, and when the curtain is retracted (ie.no outside light enters the room), the LED turns on.

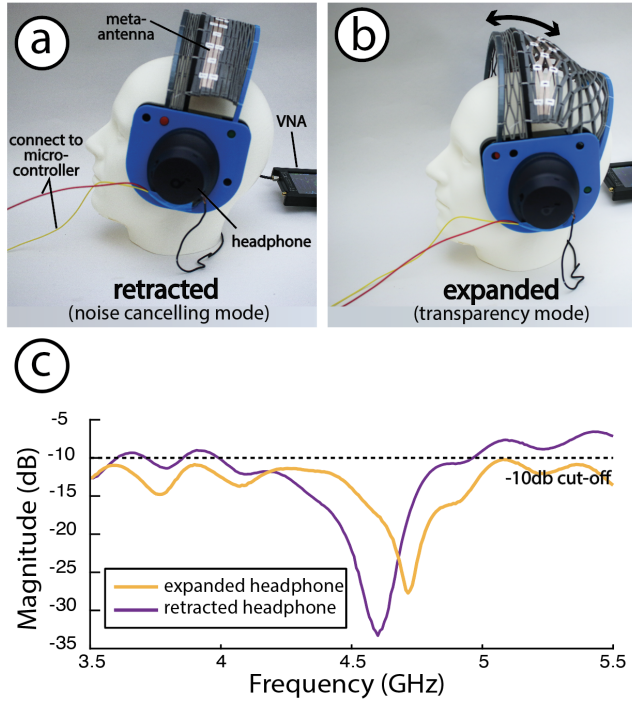


Figure 5: This shape-adaptive noise-control headphone changes from noise canceling mode to transparency mode upon expanding.

3.2 Adaptive Personal Electronic Devices

Meta-antennas can also be integrated into adaptive personal electronic devices. We demonstrate this through a smart headphone that seamlessly transitions between noise-canceling and transparent modes, while also providing a clear physical indicator of which mode it is in. To create the smart headphone, we repurpose an existing headphone using an IRFZ44N MOSFET (transistor) to enable mode switching. In our case, the switch between modes occurs when the central component of the headphone shown in Figure 5-a, identified as the Meta-antenna, expands and bends (interaction primitive). This expansion of **66.67%** alters the resonance frequency by 0.12 GHz (120 MHz), which is a resonance shift of **2.6%**. This change adjusts the transistor pin, switching the headphone mode from noise-canceling to transparent.

3.3 Interactive Tangible User Interfaces

We leverage the three distinct states generated by Meta-antenna's rotational deformation to encode meaningful interactions within

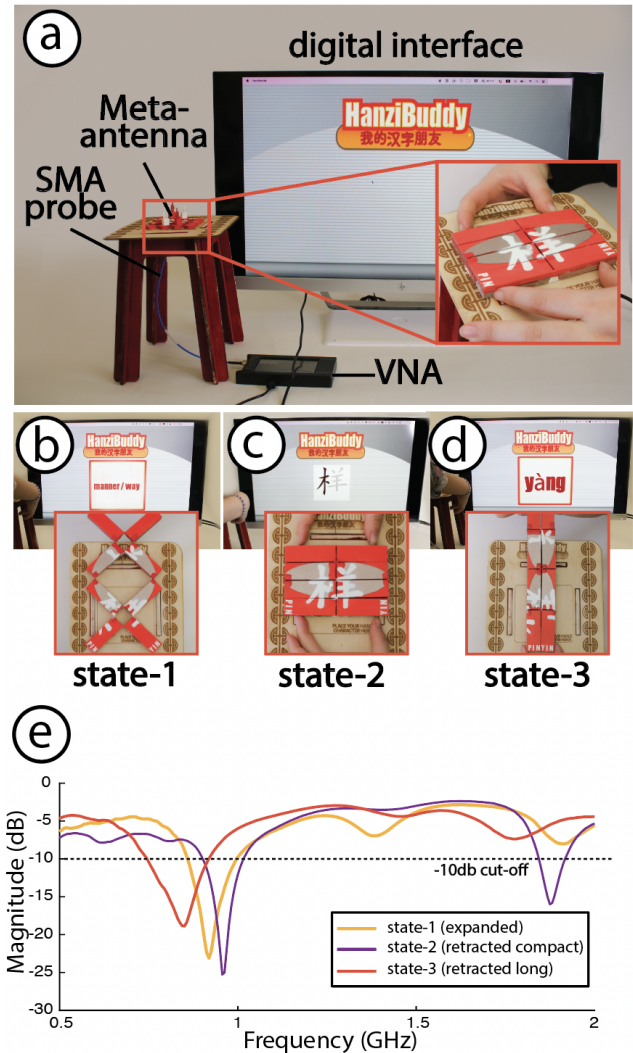


Figure 6: This Hanzi tangible learning tool reconfigures to three different Meta-antenna states.

each of its physical configuration. In this application, we demonstrate a language learning tangible that is used as a "dynamic" flash-card for learning Chinese characters (Hanzi) and interacts with a digital interface. When learning Hanzi characters, Chinese learners memorize three sets of information: the reading of the character (Pinyin), its writing, and its meaning. This three-piece information application is well-suited for Meta-antennas, which operate in up to three distinct states. To illustrate this, we embedded the information retaining the Hanzi character's meaning into State-1, its writing into State-2, and its reading (Pinyin) into State-3. We then built a Processing UI that interfaces with the Meta-antenna Hanzi as seen in Figure 6. The UI shows the character meaning (State-1), the character stroke order (State-2), and the character's reading (State-3). As Figure 6-e) shows, the PCRf for States (1-2) is 8% for an expansion rate of 193.33%, and the PCRf for States (1-3) is 4% for an expansion rate of 182.85%. The antenna is patterned with

an oval patch shape that changes pattern across states as seen in Figure 6 (b-d).

4 Conclusion

We presented Meta-antenna, a design and fabrication pipeline for creating frequency-reconfigurable antennas using a single type of mechanical metamaterial structure that can transition between three distinct geometric and frequency states. Our approach offers a straightforward method for creating mechanically reconfigurable antennas with metamaterials. Through our applications, we demonstrated how Meta-antennas can be integrated into various dynamic objects and across different scales. These applications range from responsive furniture for smart living to adaptive personal electronic devices and interactive tangible user interfaces. We believe our work demonstrates the potential of combining metamaterials and antennas in a hybridly functional way, paving the way for more means to weave antennas into the fabric of our daily lives. We hope our work inspires further exploration of alternative geometries for antennas and hybridly functional structures, like how our Meta-antenna serves as both a metamaterial and an antenna.

References

- [1] Rosette Maria Bichara, Joseph Costantine, Youssef Tawk, and Maria Sakovsky. 2023. A multi-stable deployable quadrifilar helix antenna with radiation reconfigurability for disaster-prone areas. *Nature Communications* 14, 1 (2023), 8511.
- [2] Christos G Christodoulou, Youssef Tawk, Steven A Lane, and Scott R Erwin. 2012. Reconfigurable antennas for wireless and space applications. *Proc. IEEE* 100, 7 (2012), 2250–2261.
- [3] Joseph Costantine, Youssef Tawk, Jonathan Woodland, Noah Flaum, and Christos G Christodoulou. 2014. Reconfigurable antenna system with a movable ground plane for cognitive radio. *IET Microwaves, Antennas & Propagation* 8, 11 (2014), 858–863.
- [4] Jun Gong, Olivia Seow, Cedric Honnet, Jack Forman, and Stefanie Mueller. 2021. MetaSense: Integrating Sensing Capabilities into Mechanical Metamaterial. In *The 34th Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '21). Association for Computing Machinery, New York, NY, USA, 1063–1073. <https://doi.org/10.1145/3472749.3474806>
- [5] Carla Hertleer, Annelien Van Laere, Hendrik Rogier, and Lieva Van Langenhove. 2010. Influence of relative humidity on textile antenna performance. *Textile Research Journal* 80, 2 (2010), 177–183.
- [6] Alexandra Ion, Johannes Frohnhofer, Ludwig Wall, Robert Kovacs, Mirela Alishtar, Jack Lindsay, Pedro Lopes, Hsiang-Ting Chen, and Patrick Baudisch. 2016. Metamaterial mechanisms. In *Proceedings of the 29th annual symposium on user interface software and technology*. 529–539.
- [7] Vikram Iyer, Justin Chan, and Shyamnath Gollakota. 2017. 3D Printing Wireless Connected Objects. *ACM Transactions on Graphics (TOG)*.
- [8] Daehwa Kim and Chris Harrison. 2022. EtherPose: Continuous Hand Pose Tracking with Wrist-Worn Antenna Impedance Characteristic Sensing. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 58, 12 pages. <https://doi.org/10.1145/3526113.3545665>
- [9] Daehwa Kim and Chris Harrison. 2023. Pantundefinedna: Mouth pose estimation for ar/vr headsets using low-profile antenna and impedance characteristic sensing. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 83, 12 pages. <https://doi.org/10.1145/3586183.3606805>
- [10] Mina Konaković, Keenan Crane, Bailin Deng, Sofien Bouaziz, Daniel Piker, and Mark Pauly. 2016. Beyond developable: computational design and fabrication with auxetic materials. *ACM Trans. Graph.* 35, 4, Article 89 (July 2016), 11 pages. <https://doi.org/10.1145/2897824.2925944>
- [11] Yue Li, Zhijun Zhang, Jianfeng Zheng, Zhenghe Feng, and Magdy F Iskander. 2011. A compact hepta-band loop-inverted F reconfigurable antenna for mobile phone. *IEEE transactions on antennas and propagation* 60, 1 (2011), 389–392.
- [12] Chhunheng Lor, Ratanak Phon, and Sungjoon Lim. 2024. Reconfigurable transmissive metasurface with a combination of scissor and rotation actuators for independently controlling beam scanning and polarization conversion. *Microsystems & Nanoengineering* 10, 1 (2024), 40.
- [13] Brijesh Mishra, Ramesh Kumar Verma, N Yashwanth, and Rakesh Kumar Singh. 2022. A review on microstrip patch antenna parameters of different geometry and bandwidth enhancement techniques. *International Journal of Microwave and Wireless Technologies* 14, 5 (2022), 652–673.
- [14] Ibrahim T Nassar, Harvey Tsang, Dane Bardroff, Craig P Lusk, and Thomas M Weller. 2015. Mechanically reconfigurable, dual-band slot dipole antennas. *IEEE Transactions on Antennas and Propagation* 63, 7 (2015), 3267–3271.
- [15] Seunghyun Oh, Jinkwon Tak, and Jaehoon Lee. 2019. Wireless body sensor networks based on metamaterial textiles. *Nature Electronics* 2 (2019), 140–145.
- [16] Liton Chandra Paul, Md Sarwar Hosain, Sohag Sarker, Makhluq Hossain Prio, Monir Morshed, and Ajay Krishno Sarkar. 2015. The effect of changing substrate material and thickness on the performance of inset feed microstrip patch antenna. *American Journal of Networks and Communications* 4, 3 (2015), 54–58.
- [17] Athira Raveendran, Phalguni Mathur, and Sujith Raman. 2019. Mechanically Frequency Reconfigurable Antenna and its Application as a Fluid Level Detector for Wireless Sensor Networks. In *2019 URSI Asia-Pacific Radio Science Conference (AP-RASC)*. 1–4. <https://doi.org/10.23919/URSIAP-RASC.2019.8738663>
- [18] Maria Sakovsky, Jan Negele, and Joseph Costantine. 2023. Electromagnetic Reconfiguration Using Stretchable Mechanical Metamaterials. *Advanced Science* 10, 6 (2023), 2203376. <https://doi.org/10.1002/advs.202203376> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/advs.202203376>